

Viral Diseases

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Key Words: measles, HIV, tobacco mosaic virus, vaccination side effect

DO NOW: Answer these three review questions below.

1. What did Semmelweis make doctors do and what were the results?
[2 marks]

2. Joseph Lister developed a _____.
Louis Pasteur developed v _____.
_____ were invented which helped us to see pathogens more easily. [3]

3. Describe ways to stop pathogens from spreading. [4]



Review

1. What did Semmelweis make doctors do and what were the results [2 marks]

Wash hands after handling dead bodies[1]
The number of deaths from childbed fever significantly decreased. [1]

2. Joseph Lister developed **antiseptic**. Louis Pasteur developed **vaccinations**. **Microscopes** were invented which helped us to see pathogens more easily. [3]

3. Describe ways to stop pathogens from spreading. [4]

Simple hygiene methods like keeping raw and cooked food separate, washing hands before and after using the toilet and handling food. [1]

Destroying and controlling vectors. Destroy breeding Sites for mosquitos or using mosquito nets. [1]

Isolating infected individuals [1]

Using vaccination to prevent the spread [1]

LOb: Understand three viral diseases, their symptoms and their effective treatment.

Keywords: measles, HIV, tobacco mosaic virus, vaccination side effect

Success Criteria:

- Recall key features and characteristics of a virus
- Describe and explain measles, HIV and tobacco mosaic virus



LATIN

Did you know... the name virus was coined from the Latin for slimy liquid or poison.

Complete these activity boxes for what you have learnt

Questions



1. State three methods of preventing spread of disease.
2. Why does isolating individuals stop spread of disease?

True or false



1. A vector is something that carries disease
2. Joseph Lister created vaccinations
3. Antiseptic destroy pathogens outside the body
4. Ebola is non communicable

Expand sentence



A vaccine contains

The discovery of the microscope was important because...

Odd one out



strong

pathogen

Jenner

memory

immune

injection

Review

Questions



1. State three methods.
Vaccine, isolation, hygiene
2. Isolation prevents the pathogen from getting close to another person and infecting them

True or false



1. A vector is something that carries disease
2. Joseph Lister created vaccinations
3. Antiseptic destroy pathogens outside the body
4. Ebola is non communicable

Expand sentence



A vaccine contains a weakened form of the pathogen so it does not harm us.

The discovery of the microscope was important because it showed us pathogens and proved that they cause disease...

Odd one out



strong

pathogen

Jenner

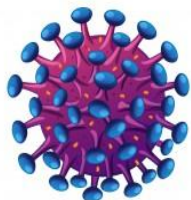
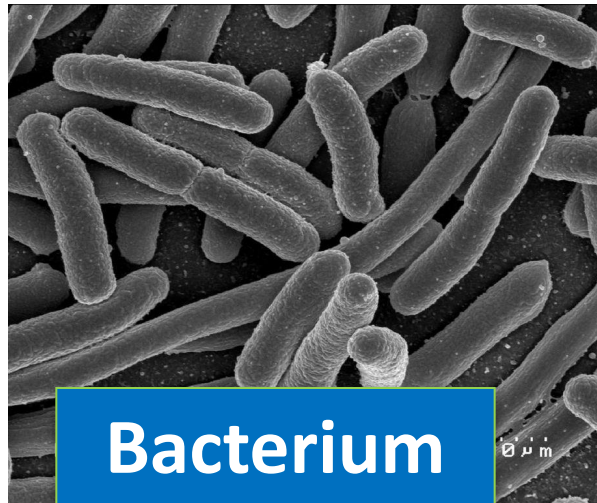
memory

immune

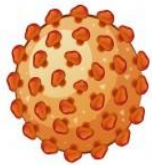
injection

All communicable diseases are caused by pathogens.

Pathogen: Microorganisms that causes disease



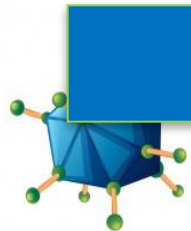
HIV



Hepatitis B



Ebola Virus



Adenovirus



Influenza



Bacteriophage

Virus



Fungus



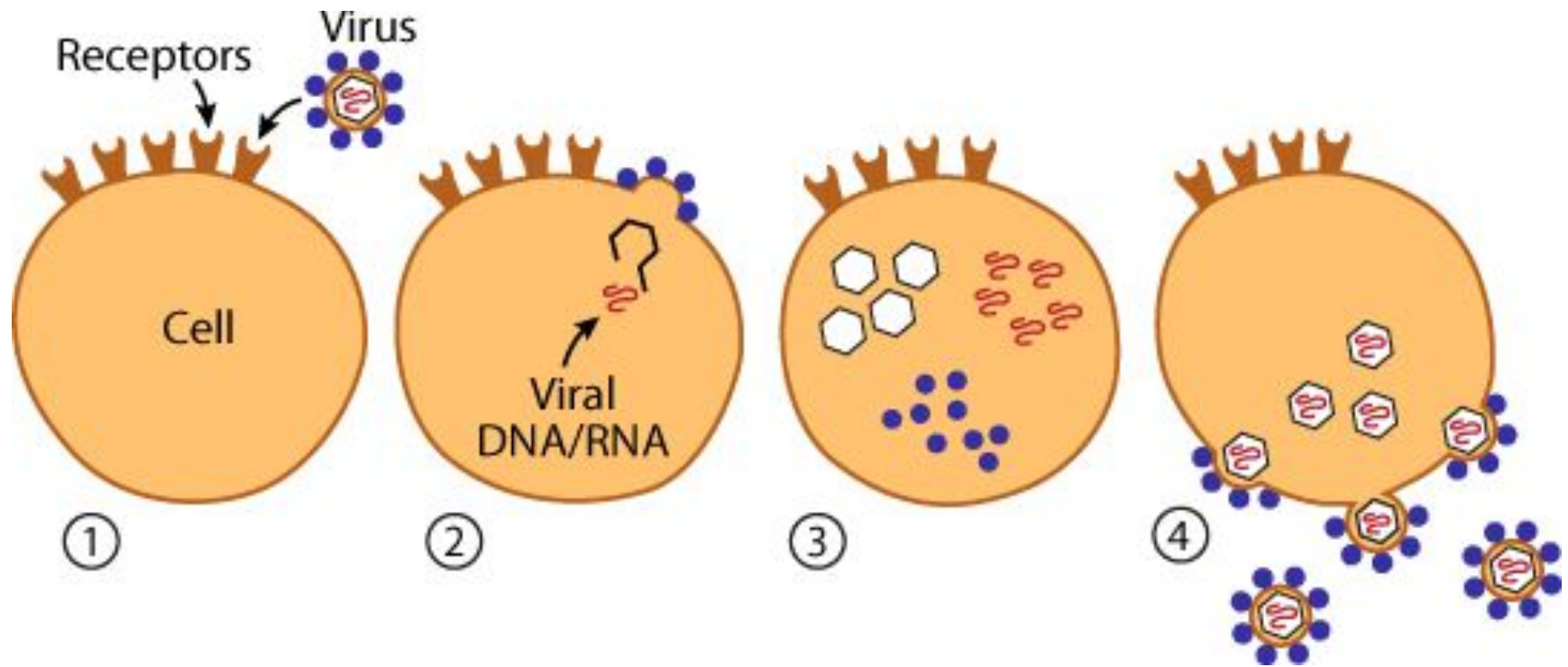
**What can you recall
about**



viruses?

It is harder to treat viral infections because viruses live inside of cells.

Antibiotics don't work



keyfacts



Viruses are not alive



Smaller than bacteria



They need a cell to reproduce



They damage cells which makes us feel ill

For these reasons we
focus on prevention



LOb: Understand three viral diseases, their symptoms and their effective treatment.

Keywords: measles, HIV, tobacco mosaic virus, vaccination

Success Criteria:

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LATIN

Did you know... the name virus was coined from the Latin for slimy liquid or poison.

Activity: Measles is a viral disease. Watch the video and complete the sheet.

WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

POWER!



Review

Symptoms



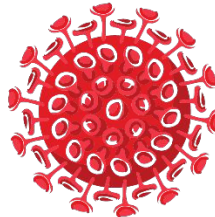
Red skin rash



fever

Pathogen

virus



Prevention

☐ Vaccinate population

☐ Isolate infected individual



Measles

Spread

Inhaling droplets



LOb: Understand three viral diseases, their symptoms and their effective treatment.

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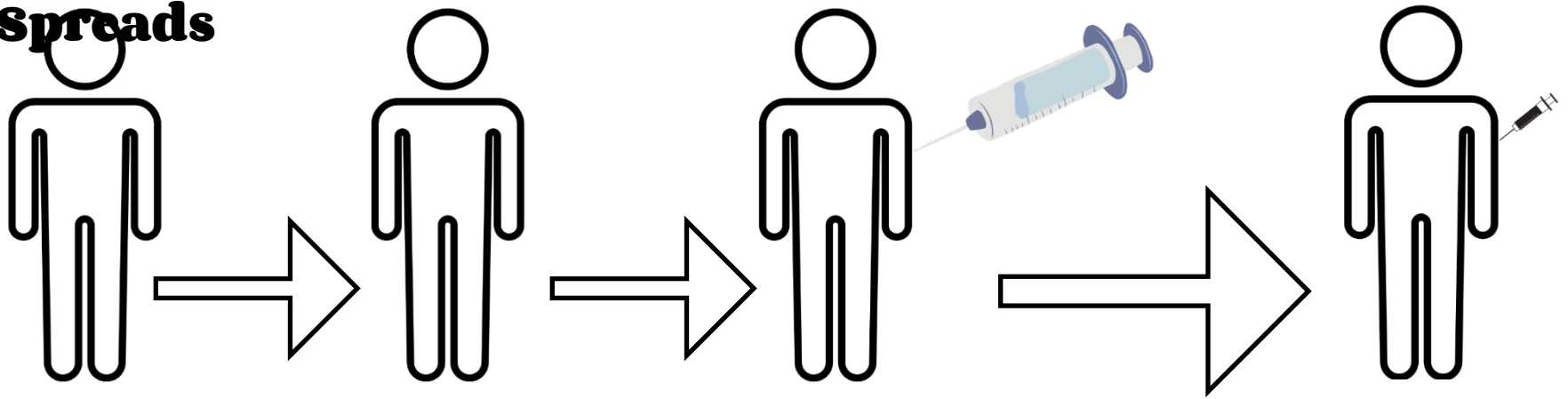
LATIN

Did you know... the name virus was coined from the Latin for slimy liquid or poison.

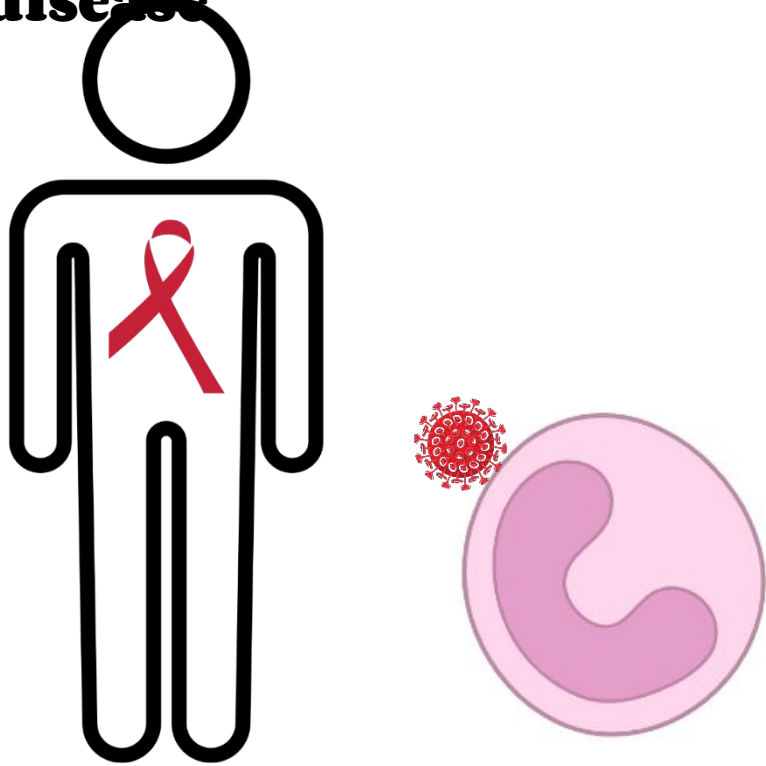
Activity: Watch the video and complete the diagram about HIV as you watch.



How it Spreads



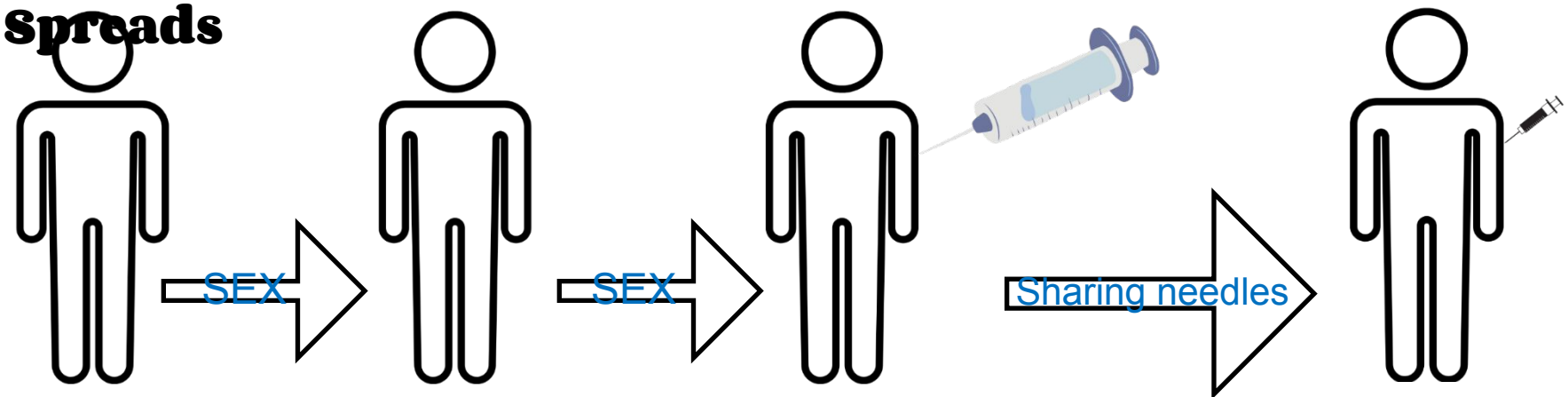
How it causes disease



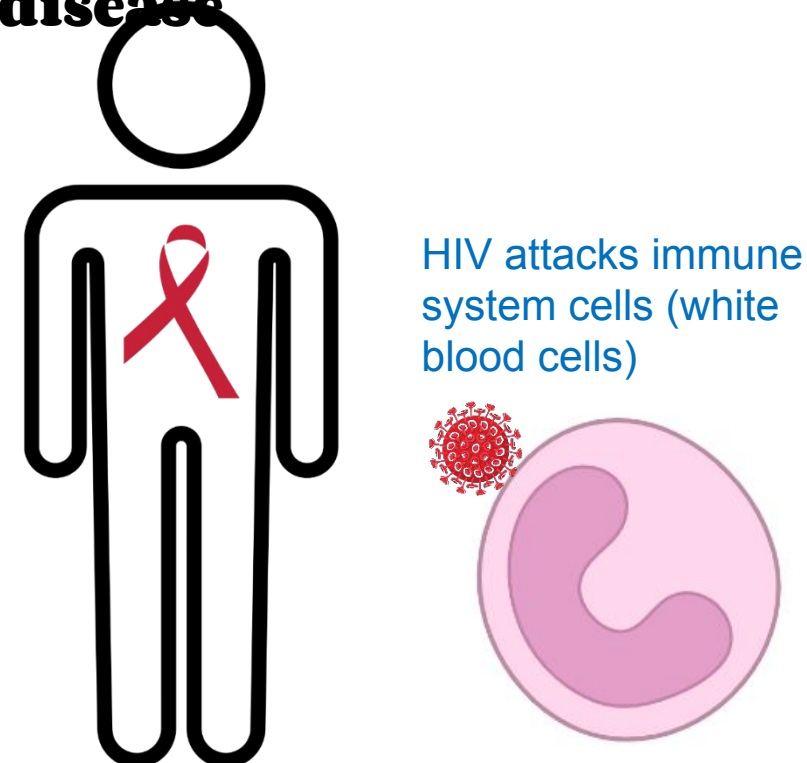
Prevention + treatment



How it Spreads



How it causes disease



When enough cells are damaged the infected person can get AIDS

Prevention + treatment

Use condoms

Anti-retroviral drugs

Use clean needles

Make it harder for your immune system fight others diseases. Patients can die of a simple flu or infection

This block contains icons for prevention and treatment: a pack of condoms, two pills (one blue and one yellow), and a syringe.

LOb: Understand three viral diseases, their symptoms and their effective treatment.

Keywords: measles, HIV, tobacco mosaic virus, vaccination side effect

Success Criteria:

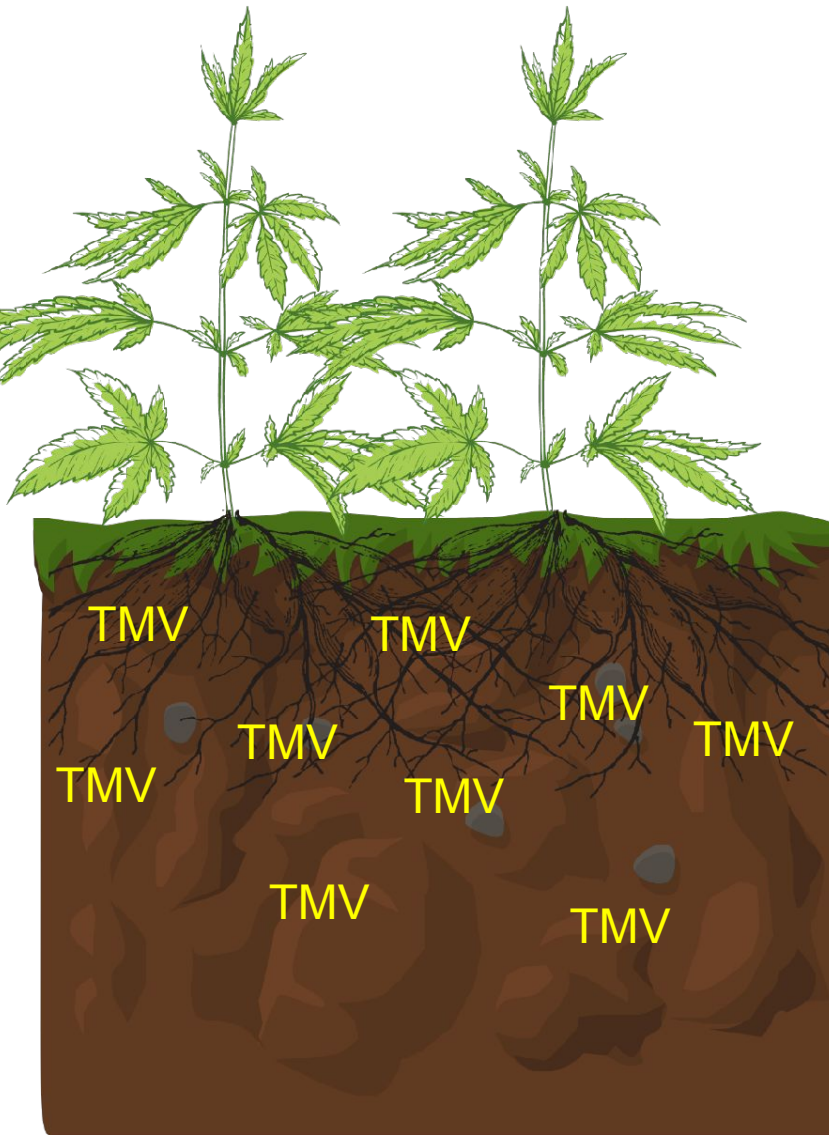
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LATIN

Did you know... the name virus was coined from the Latin for slimy liquid or poison.

Tobacco mosaic virus is a virus that attacks the leaves of tobacco plants. It doesn't usually kill them but makes leaves **discoloured**.



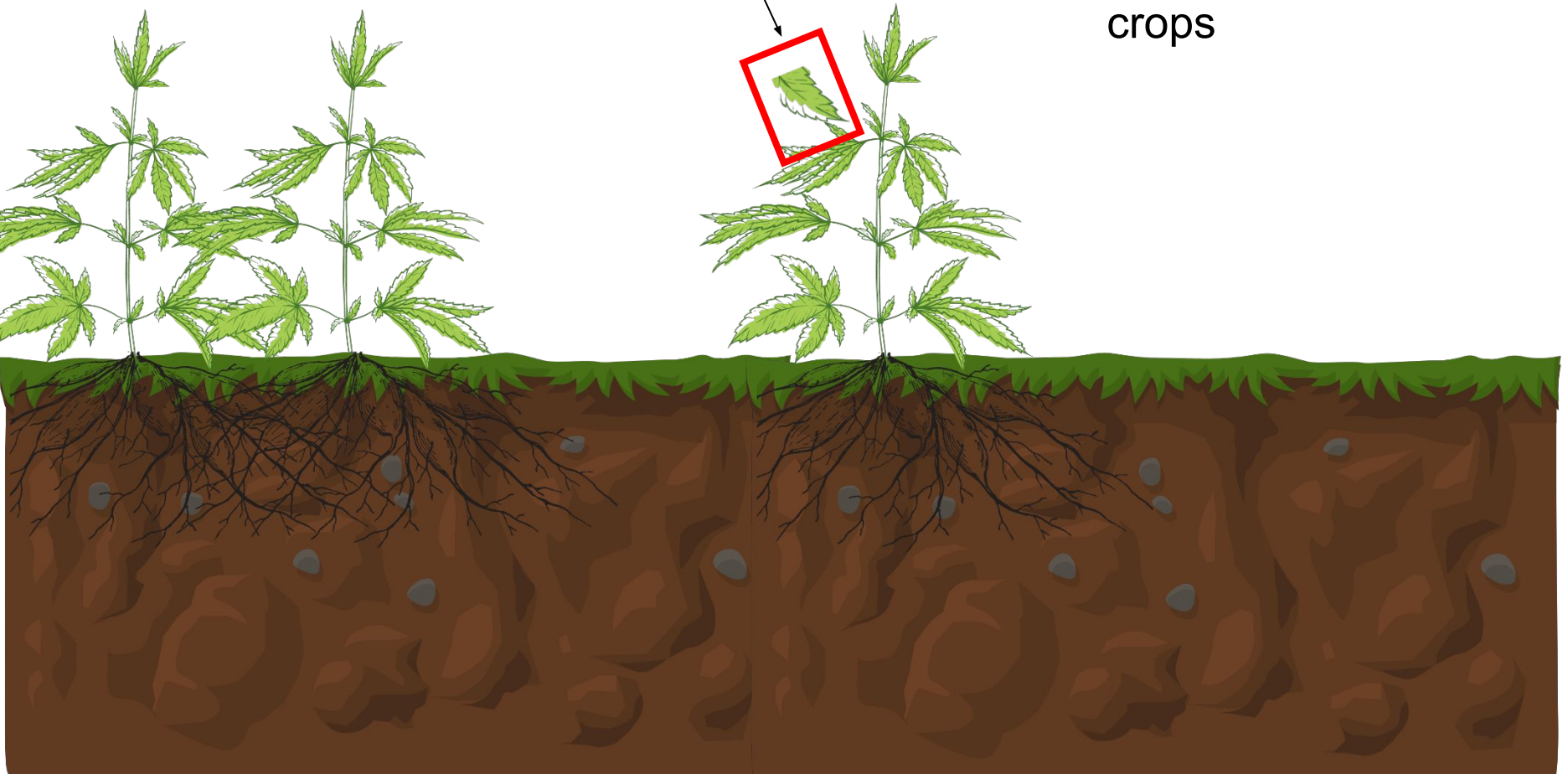
TMV in the soil is absorbed into the root systems of plants.

If a leaf infected with **TMV** touches another plant not infected then the **direct contact** will infect the new plant.

Preventi **on**

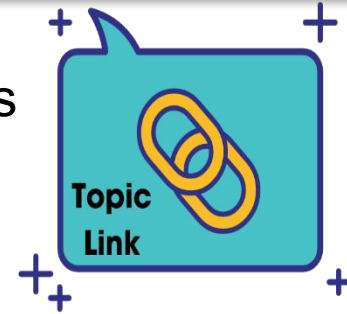
- ☐ Remove/destroy all infected plants
- ☐ Grow TMV resistant crops

TMV infect leaf
touching another plant



TMV doesn't kill plants but how will leaves like this disadvantage the plant?

1. Reduction in chlorophyll = reduction in rate of photosynthesis
2. Less glucose produced
3. Plant is weakened, growth reduced = less crop yield



Bioenergetics



Activity: Complete the sheet questions.



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LATIN

Did you know... the name virus was coined from the Latin for slimy liquid or poison.

Does vaccination for measles work?

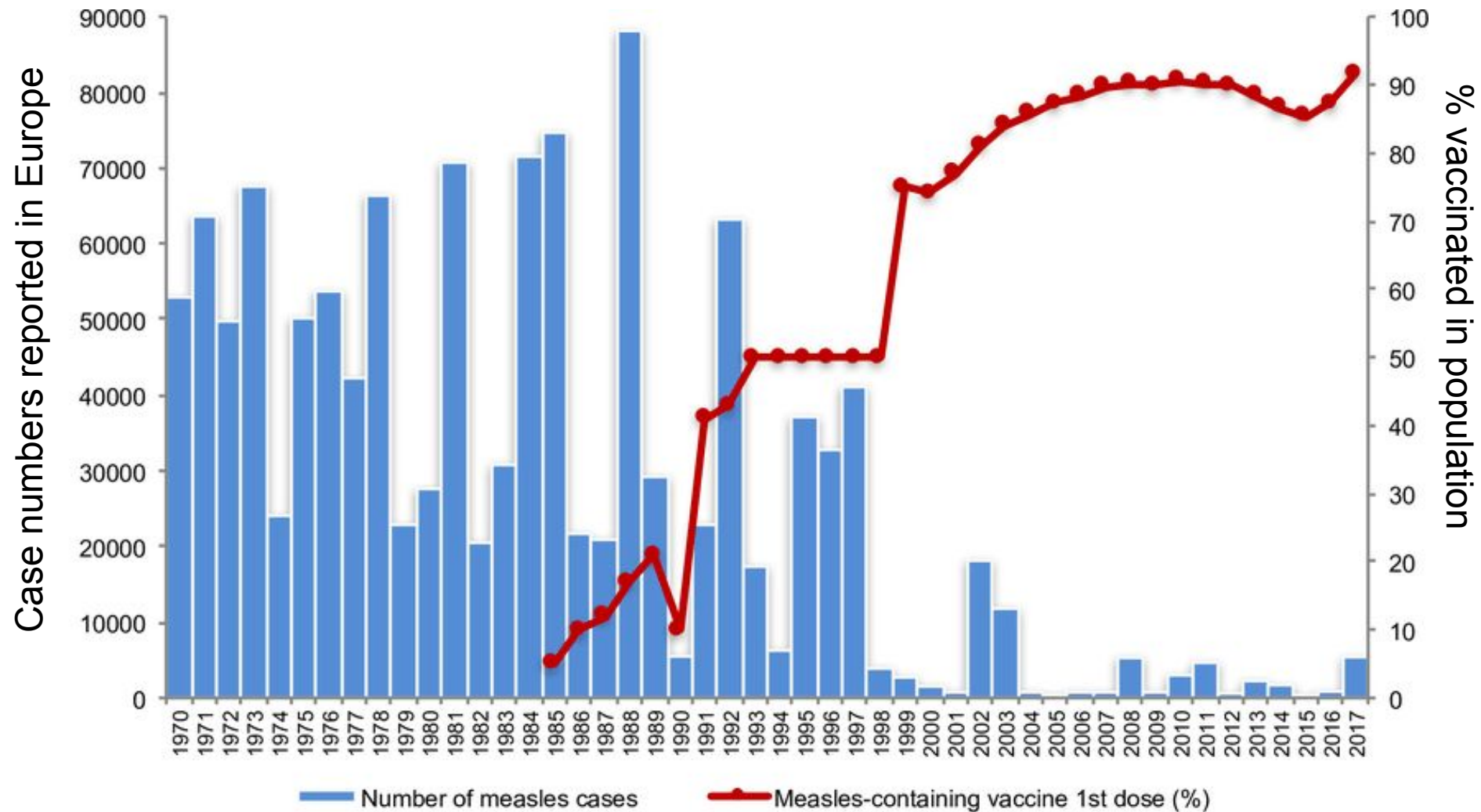
Between 1970 -1984
no vaccination
but a high measles rate

Vaccine uptake increases,
measles cases start to
decrease

High vaccine uptake
low measles cases



Review



keyfacts

 **Antibiotics** are a drug that kills bacteria. They **don't work on viruses** because viruses live inside cells.

 **Measles**: symptoms= **red rash**, fever. How it spreads: **droplets** in the air. Prevention: **Vaccination**, isolate individual



 **HIV**: symptoms= fever. How it spreads: **exchange of bodily fluids**. Prevention: antiviral drugs, **condoms**, **clean needles**

 **TMV**: symptoms= **discolouration** of leaves. How it spreads: **direct contact**. Prevention: **remove** infected plants

Plenary: Word match

HIV

Measles

TMV

isolation

Bodily fluids

Young children

Remove

Fever

Needle
sharing

Direct contact

rash

condom

Resistant crops

Droplets

vaccination

Plenary

Complete the all questions in this learning triangle

